Outlier Detection and Removal Analysis

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OUTLIER DETECTION ANALYSIS
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Z-SCORE THRESHOLD: 2.0
DETECTION METHODS: Multiple (Z-score, Modified Z-score, IQR, Extreme values)
OUTLIERS DETECTED: 2

OUTLIER DETAILS:
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• 30%M,10%C (Votes: 3):
  Final Accuracy: 89.87%
  Final Loss: 3.0597
  Final F1-Score: 0.900
  Standard Z-Scores:
    - Accuracy: 2.03
    - Loss: 0.82
    - F1: 1.90
  Modified Z-Scores:
    - Accuracy: 4.40
    - Loss: 0.86
    - F1: 3.75
  Other flags:
    - IQR outlier: True
    - Extreme outlier: False
  Detection reasons: Z-score: Acc=2.03, Loss=0.82, F1=1.90; Mod-Z: Acc=4.40, Loss=0.86, F1=3.75; IQR method

• 40%M,0%C (Votes: 2):
  Final Accuracy: 90.14%
  Final Loss: 3.3763
  Final F1-Score: 0.902
  Standard Z-Scores:
    - Accuracy: 1.46
    - Loss: 2.22
    - F1: 1.49
  Modified Z-Scores:
    - Accuracy: 3.33
    - Loss: 2.09
    - F1: 3.04
  Other flags:
    - IQR outlier: True
    - Extreme outlier: False
  Detection reasons: Z-score: Acc=1.46, Loss=2.22, F1=1.49; IQR method

DETECTION METHODS:
1. Standard Z-score: |Z| > 2.0
2. Modified Z-score (MAD-based): |Z| > 3.5
3. IQR method: outside Q1-1.5*IQR to Q3+1.5*IQR
4. Extreme values: >3 standard deviations

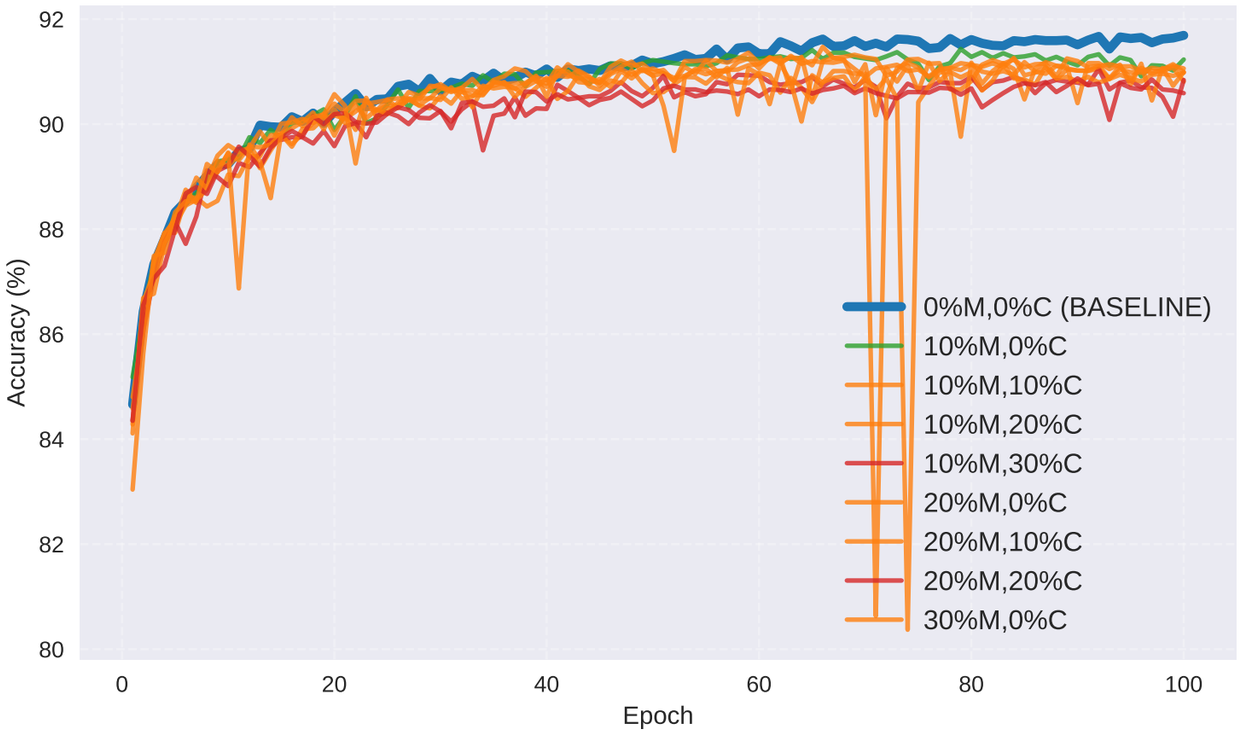
VOTING SYSTEM:
An experiment is flagged as outlier if it receives
≥2 votes from the detection methods.
Extreme outliers get extra weight (2 votes).

SAFETY LIMITS:
Maximum 50% of experiments can be removed
to prevent over-aggressive filtering.

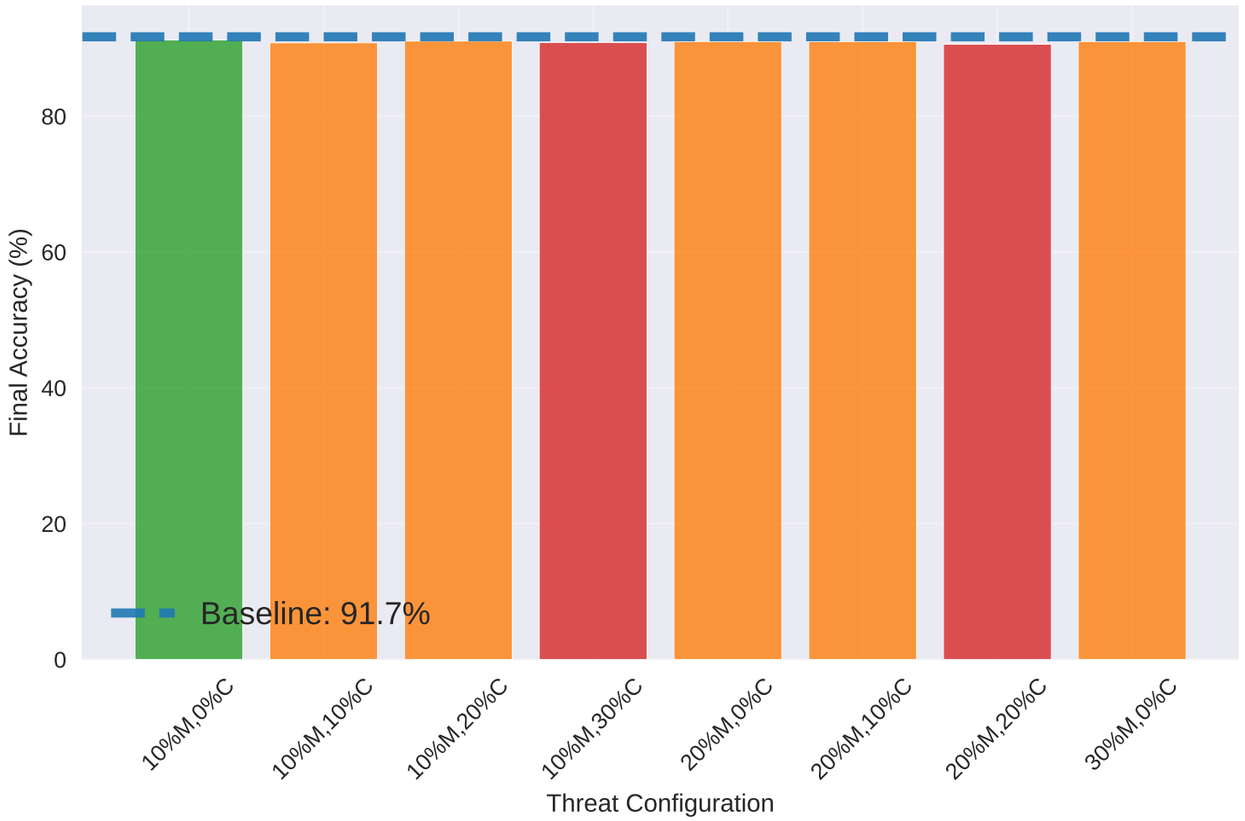
IMPACT ON ANALYSIS:
Outliers have been removed to ensure more
reliable comparisons and trend analysis.
This improves the quality of insights
derived from the FL performance data.
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FL Accuracy Analysis - Trust Experiments

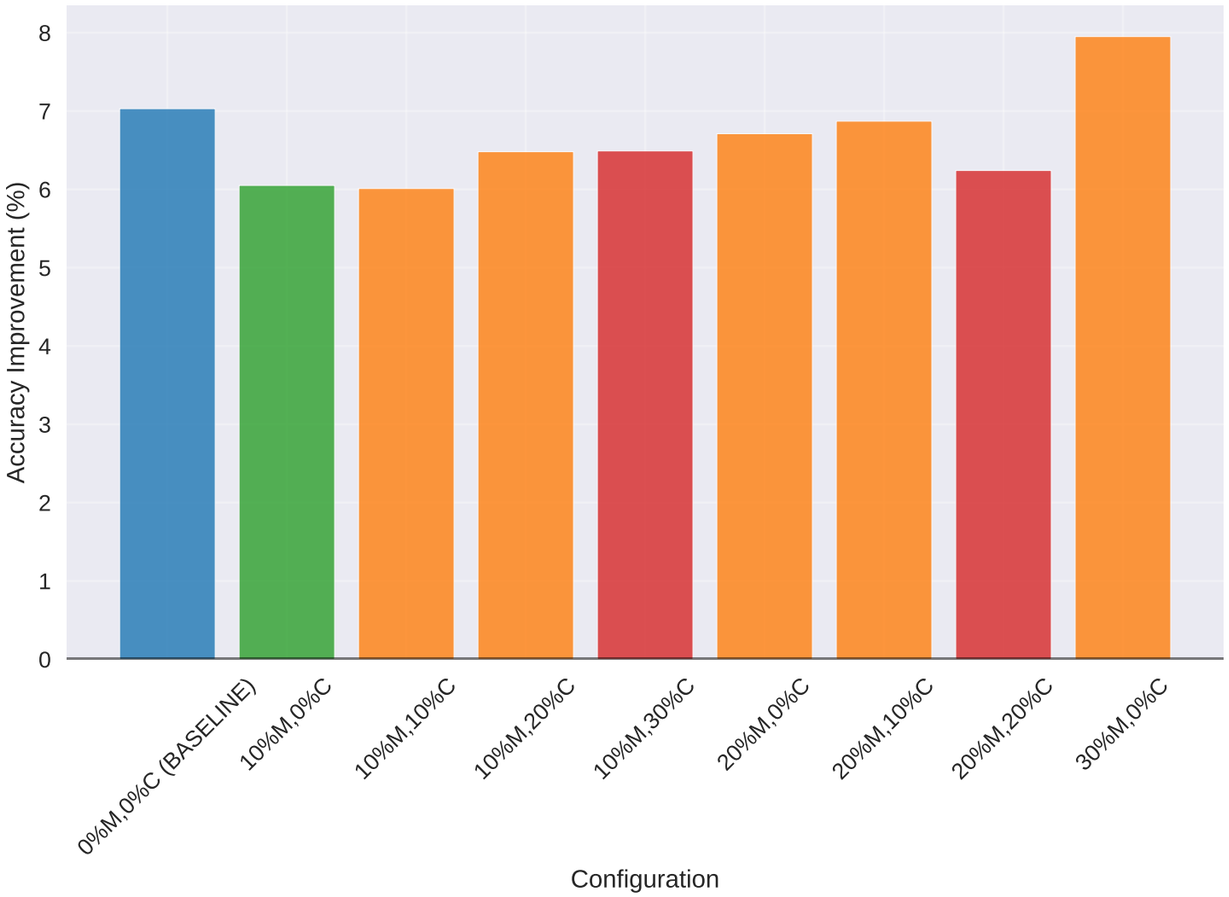
□ Accuracy Evolution Comparison



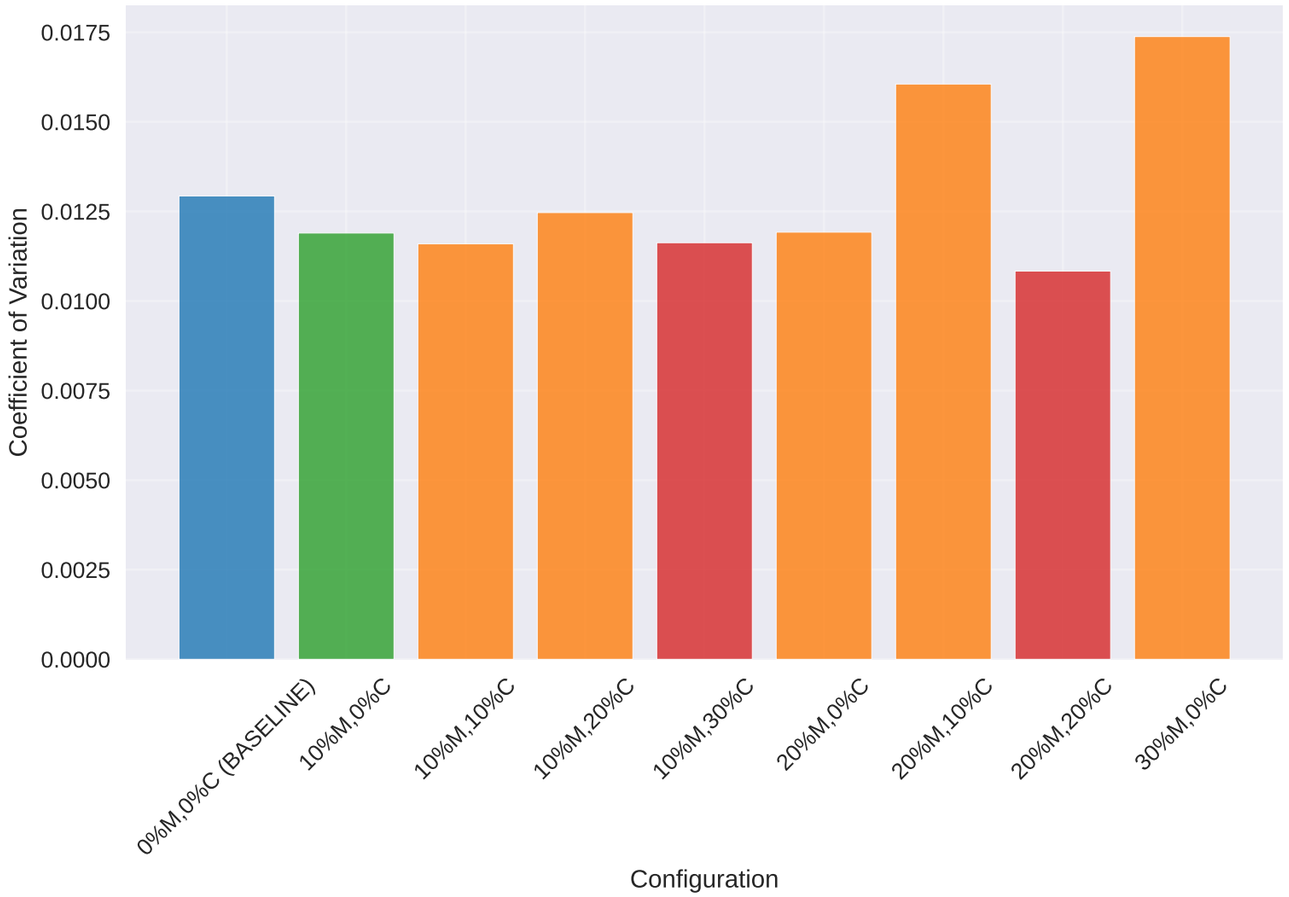
□ Final Accuracy vs Threat Level



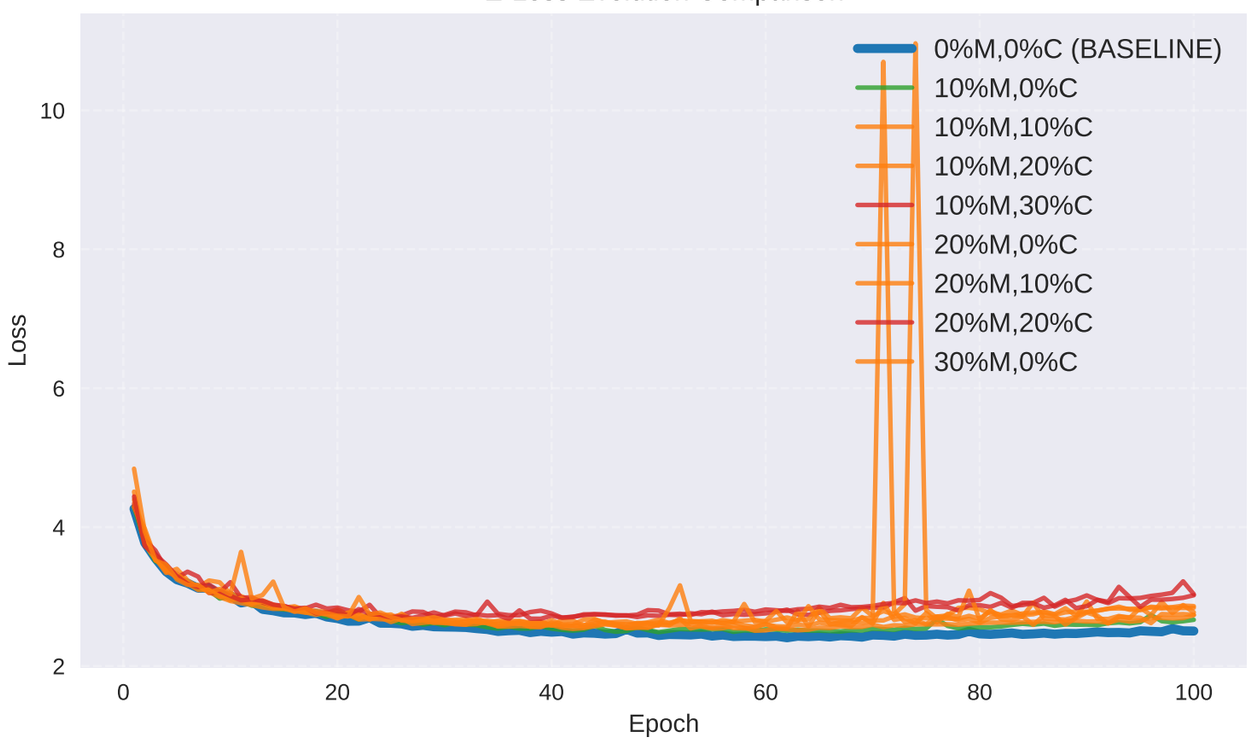
□ Learning Progress Comparison

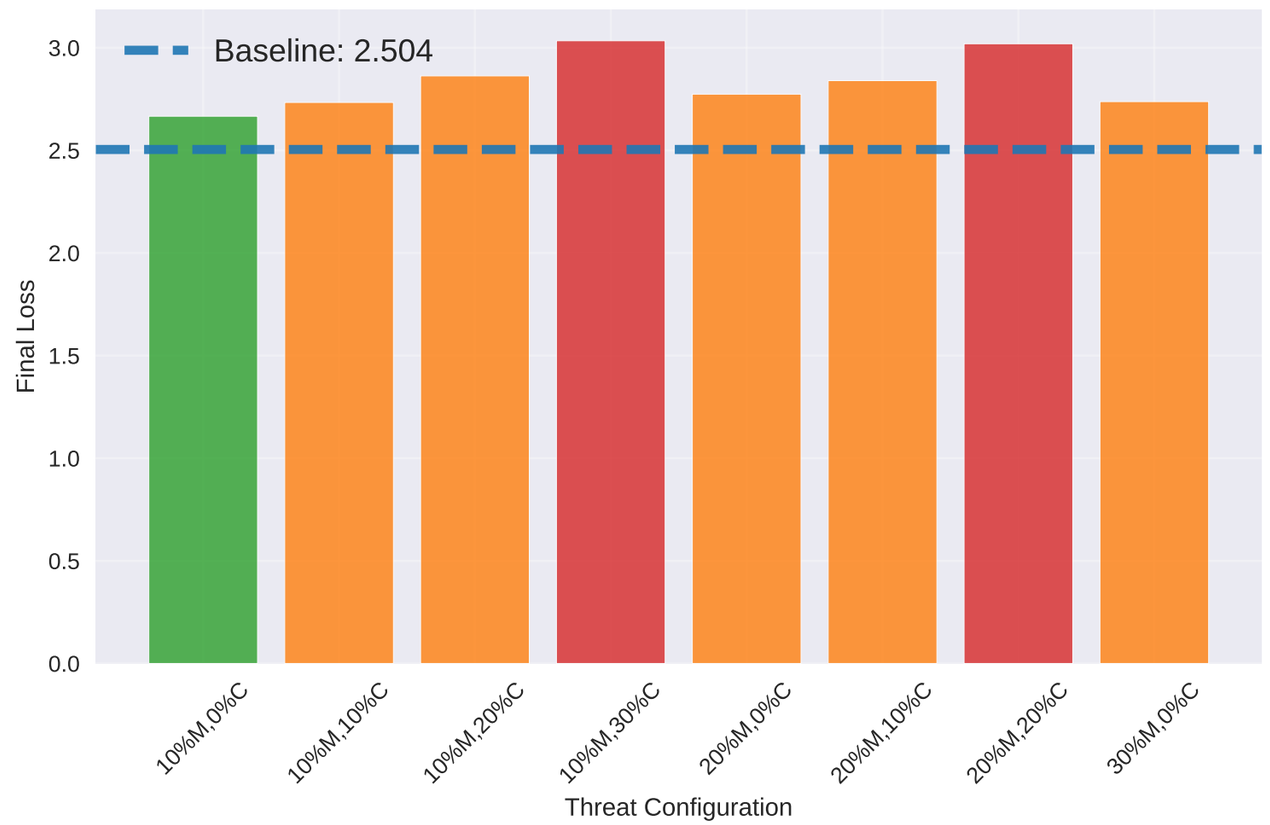


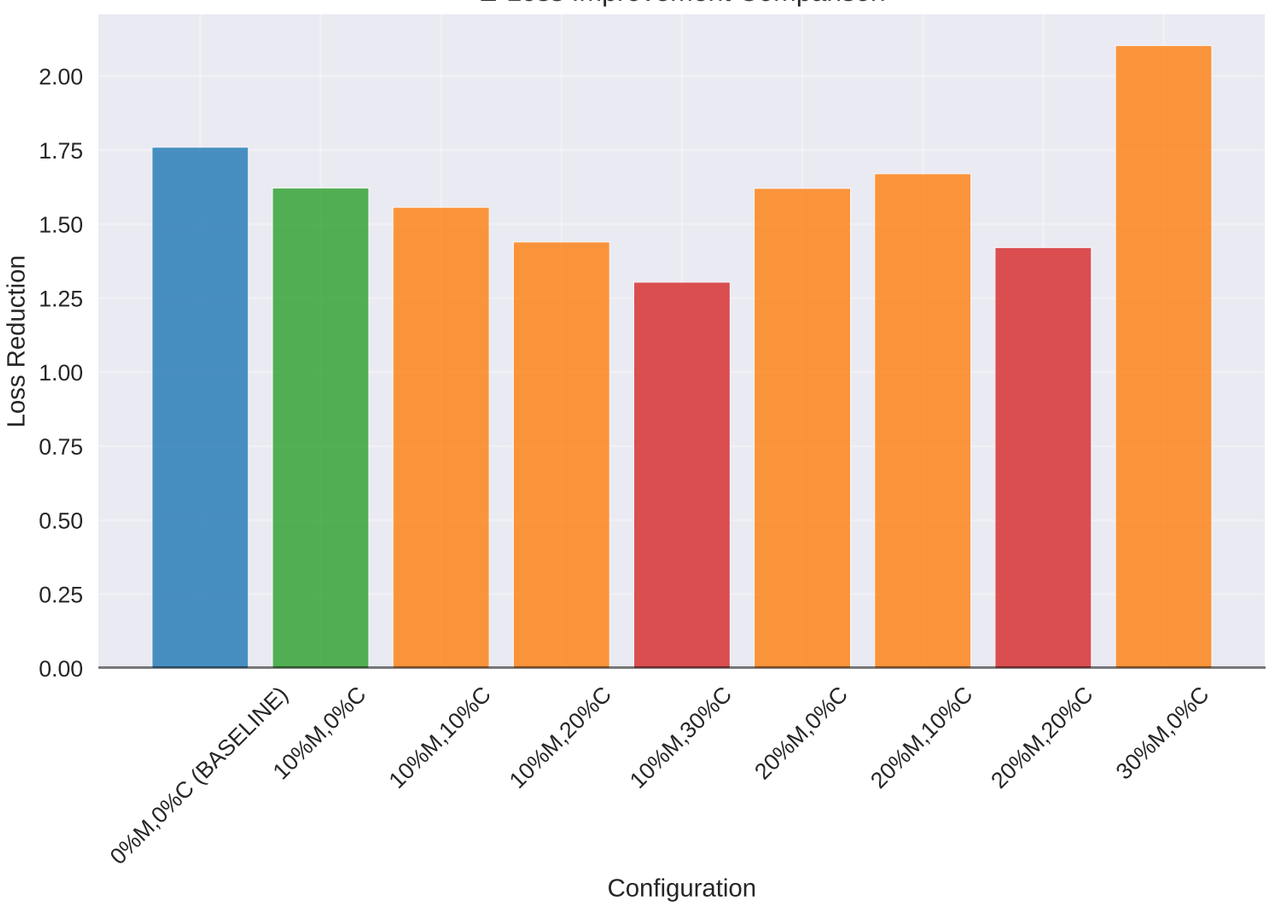
□ Training Stability Comparison

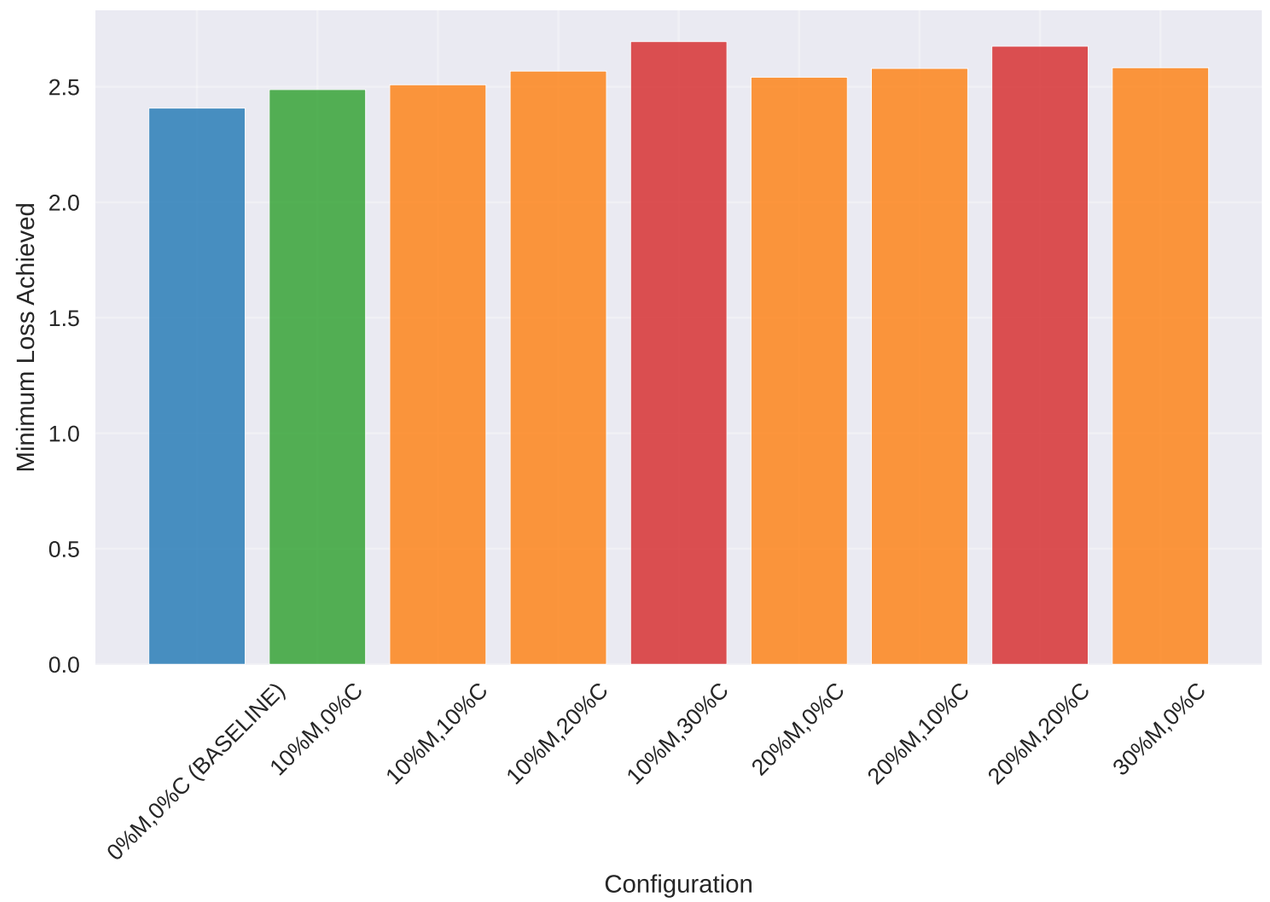


FL Loss Analysis - Trust Experiments

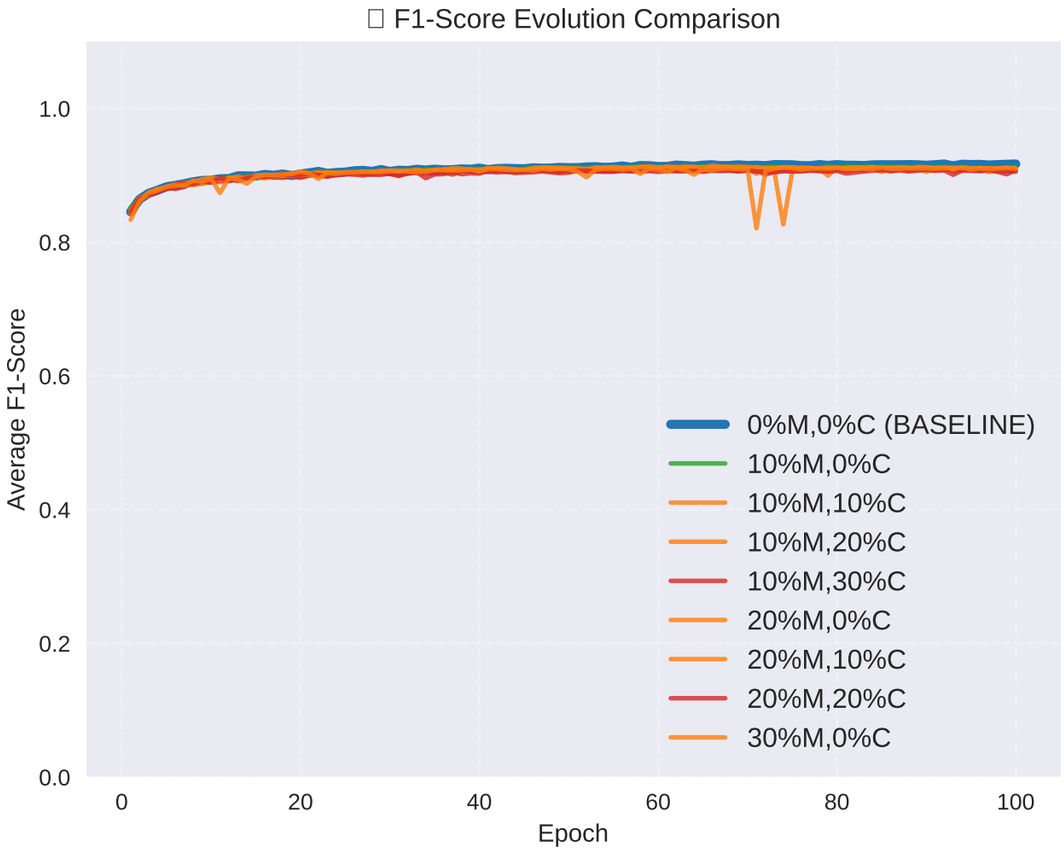


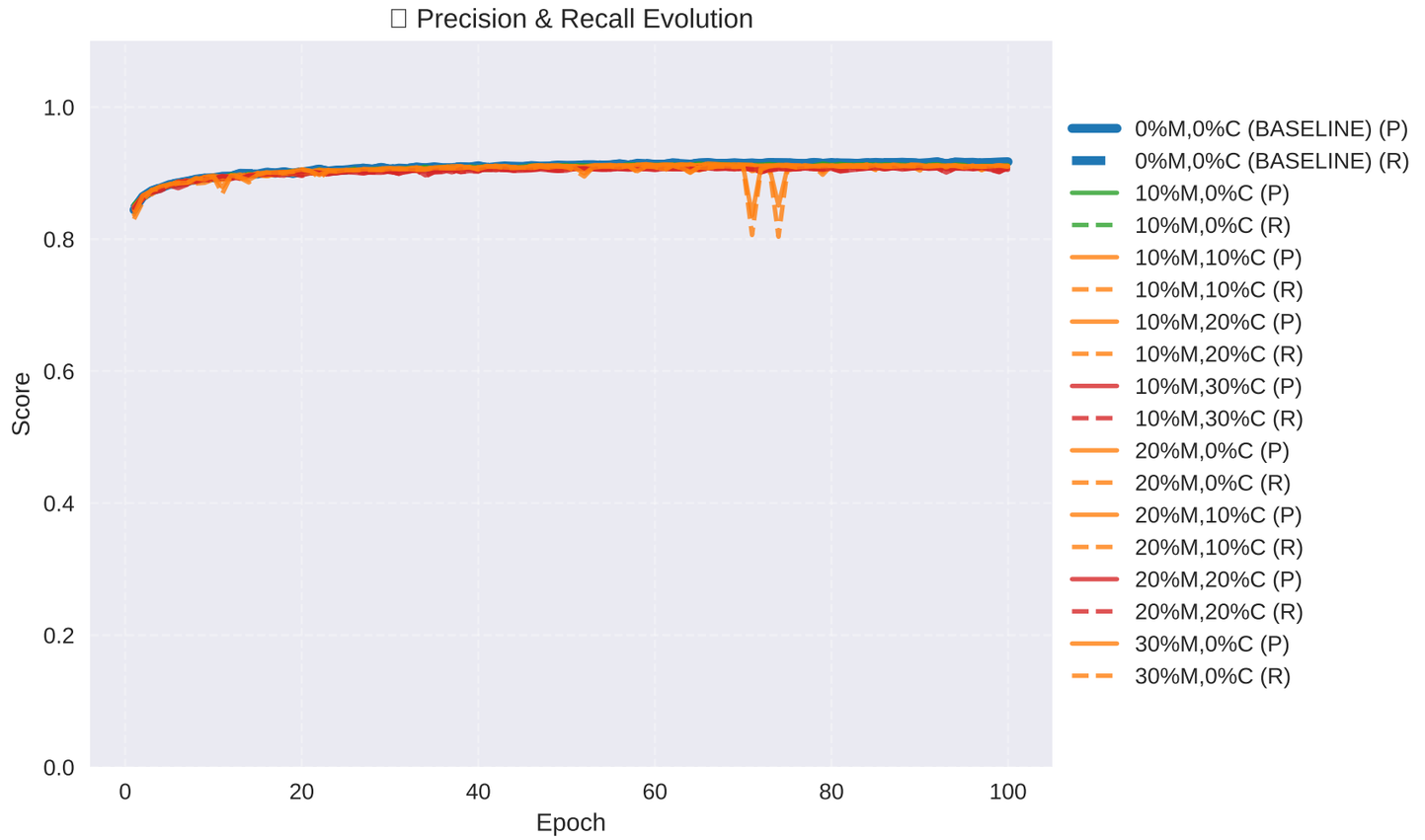


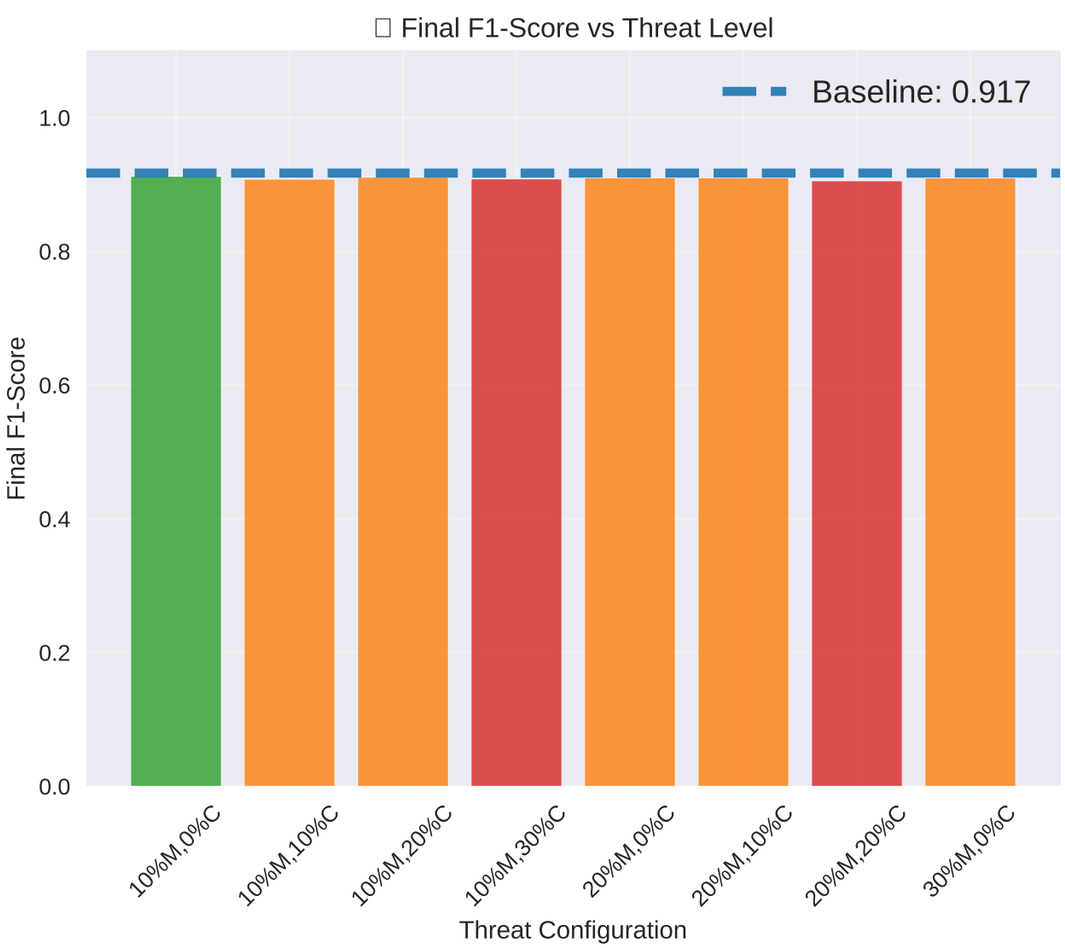


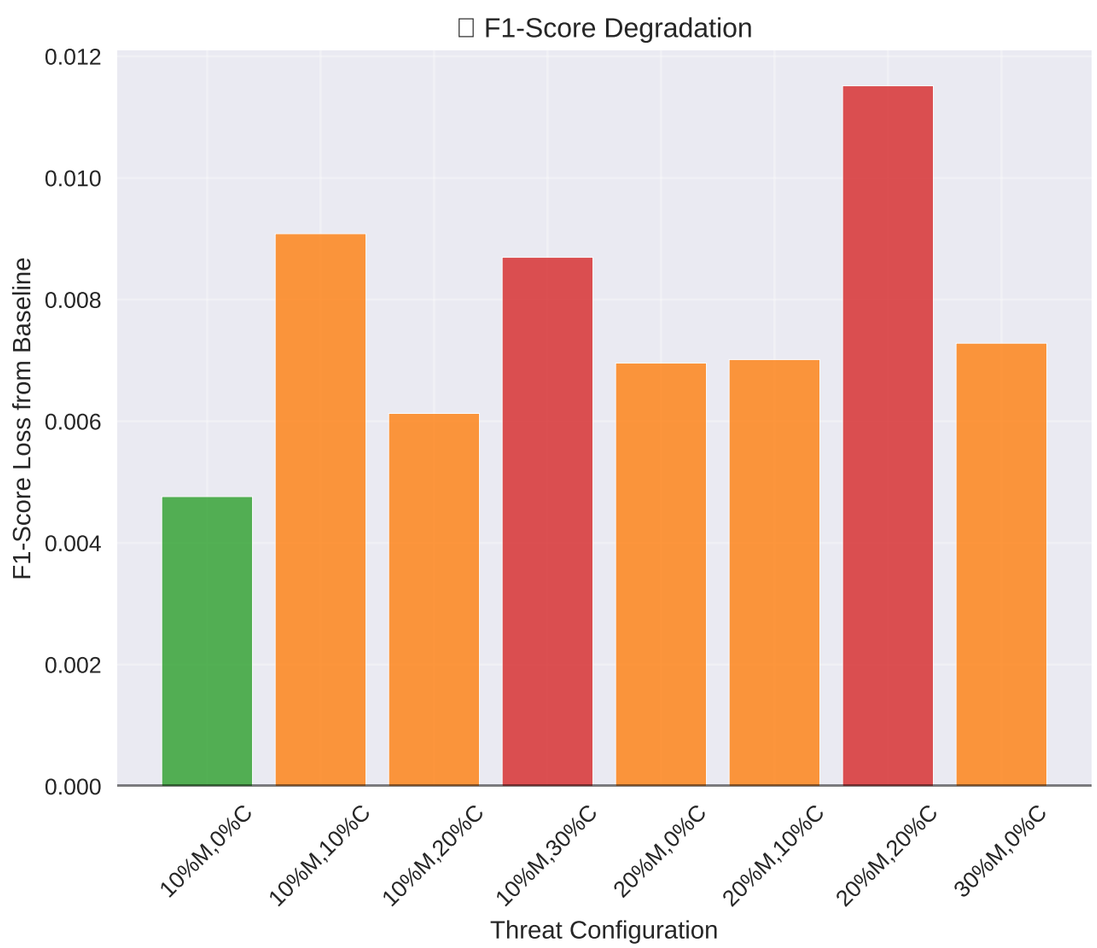


FL F1-Score Analysis - Trust Experiments

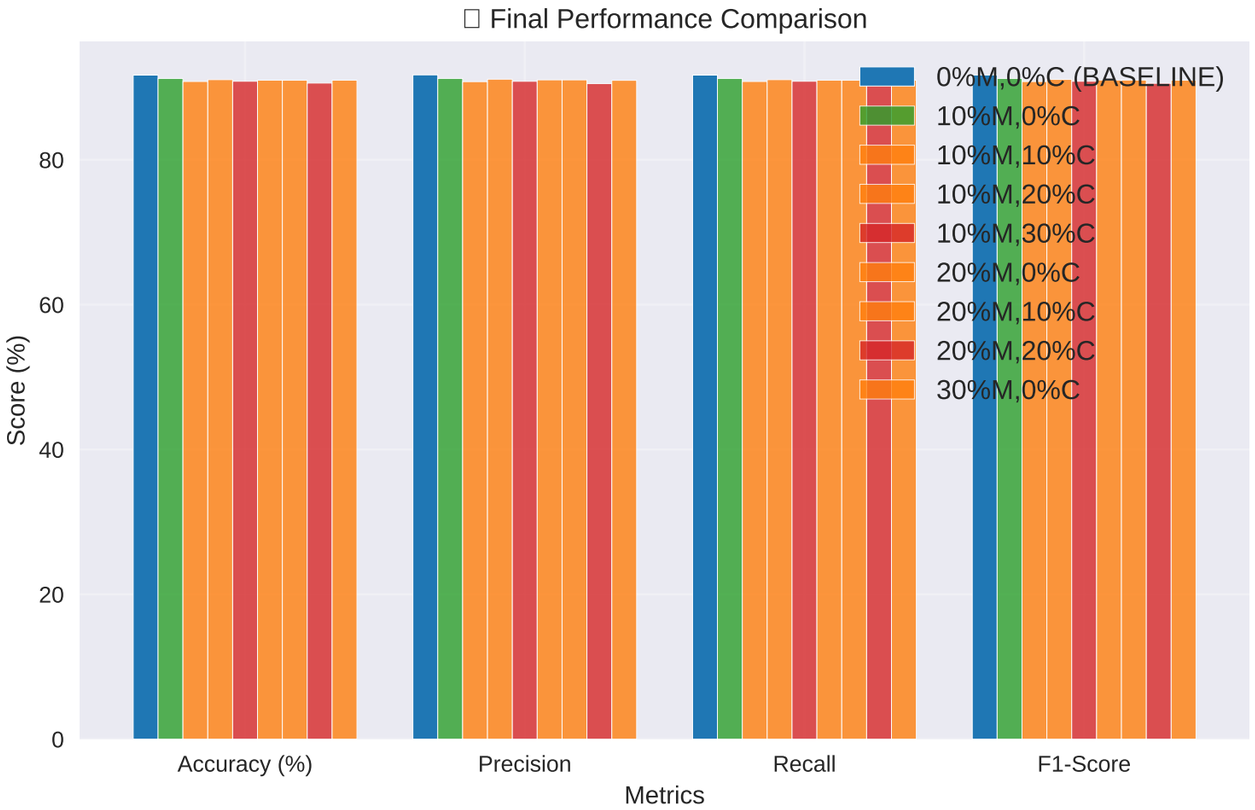


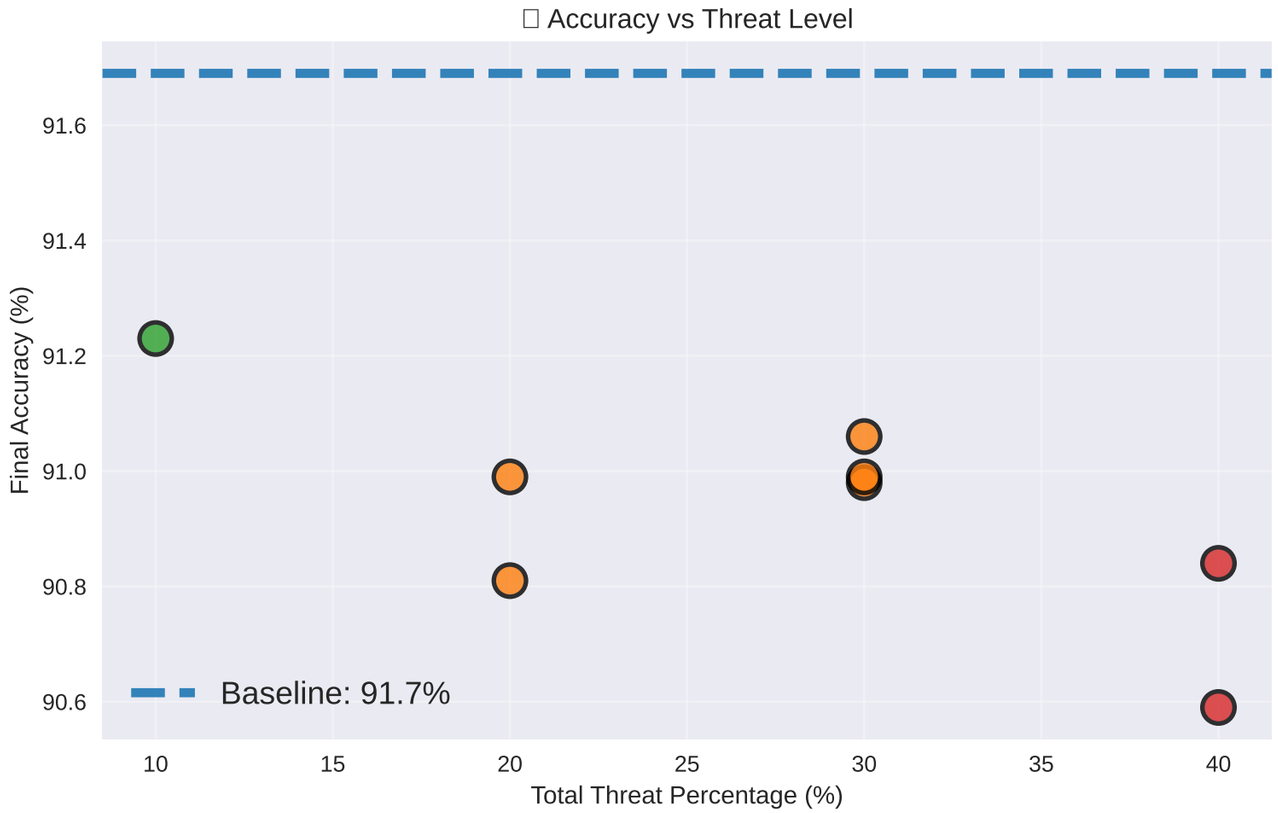


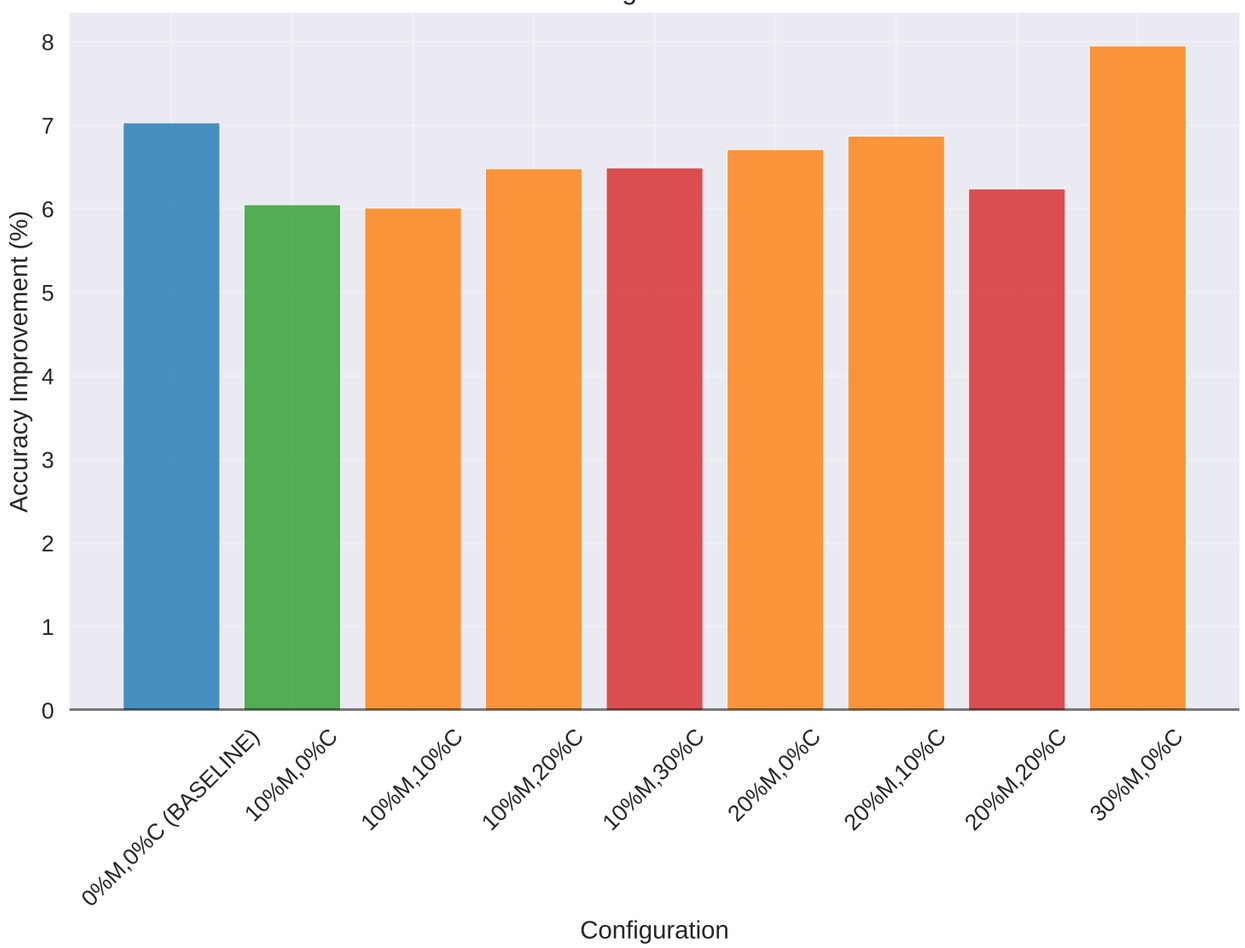


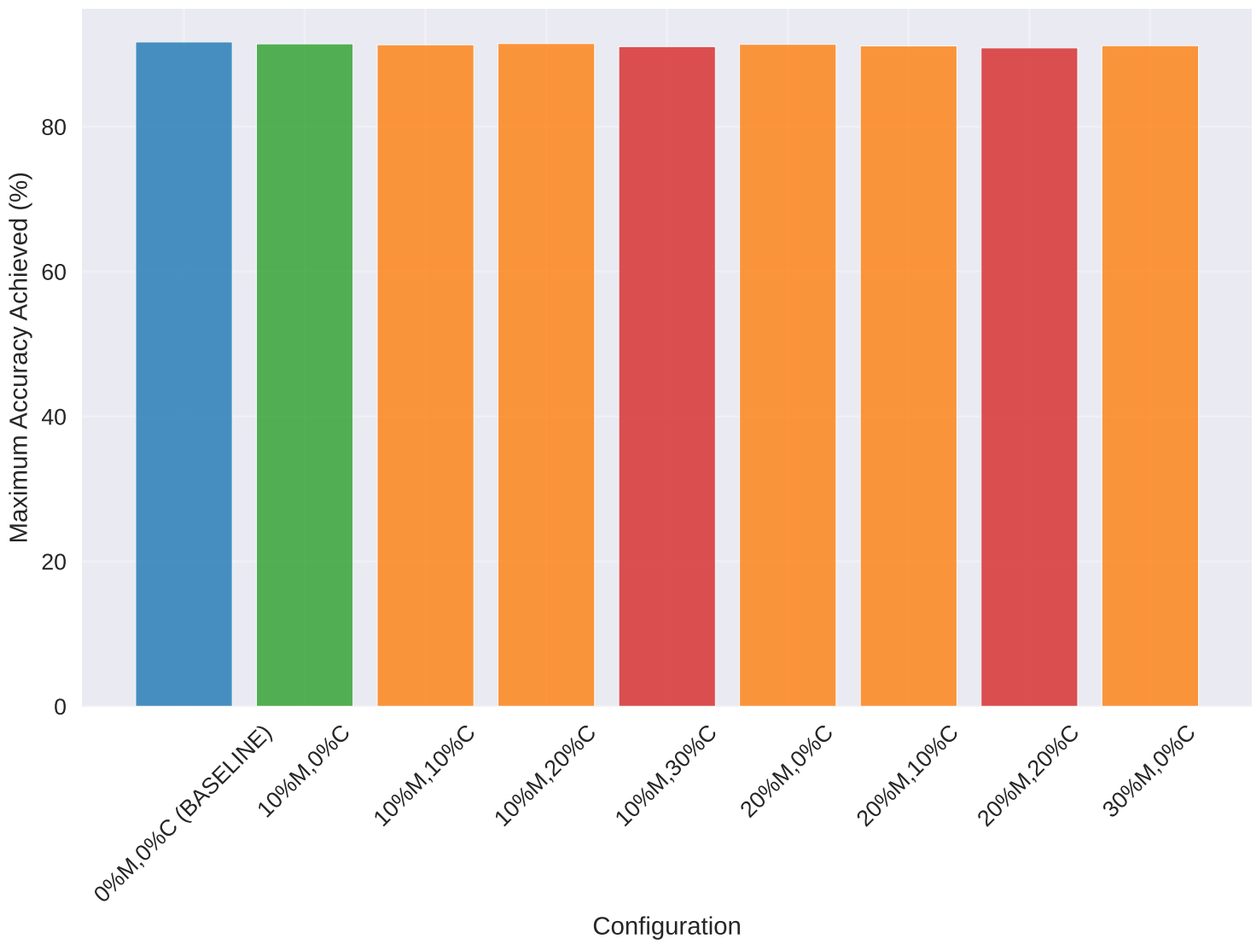


Final Performance Analysis

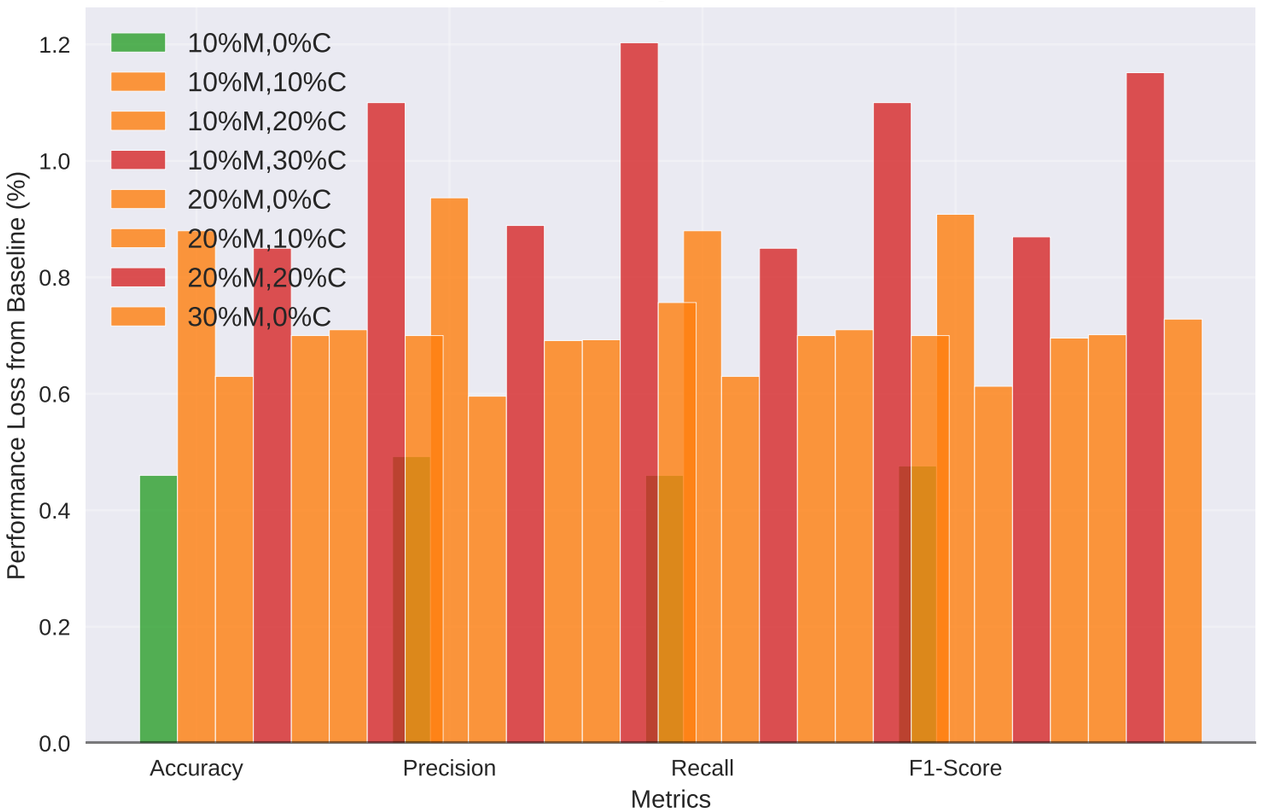


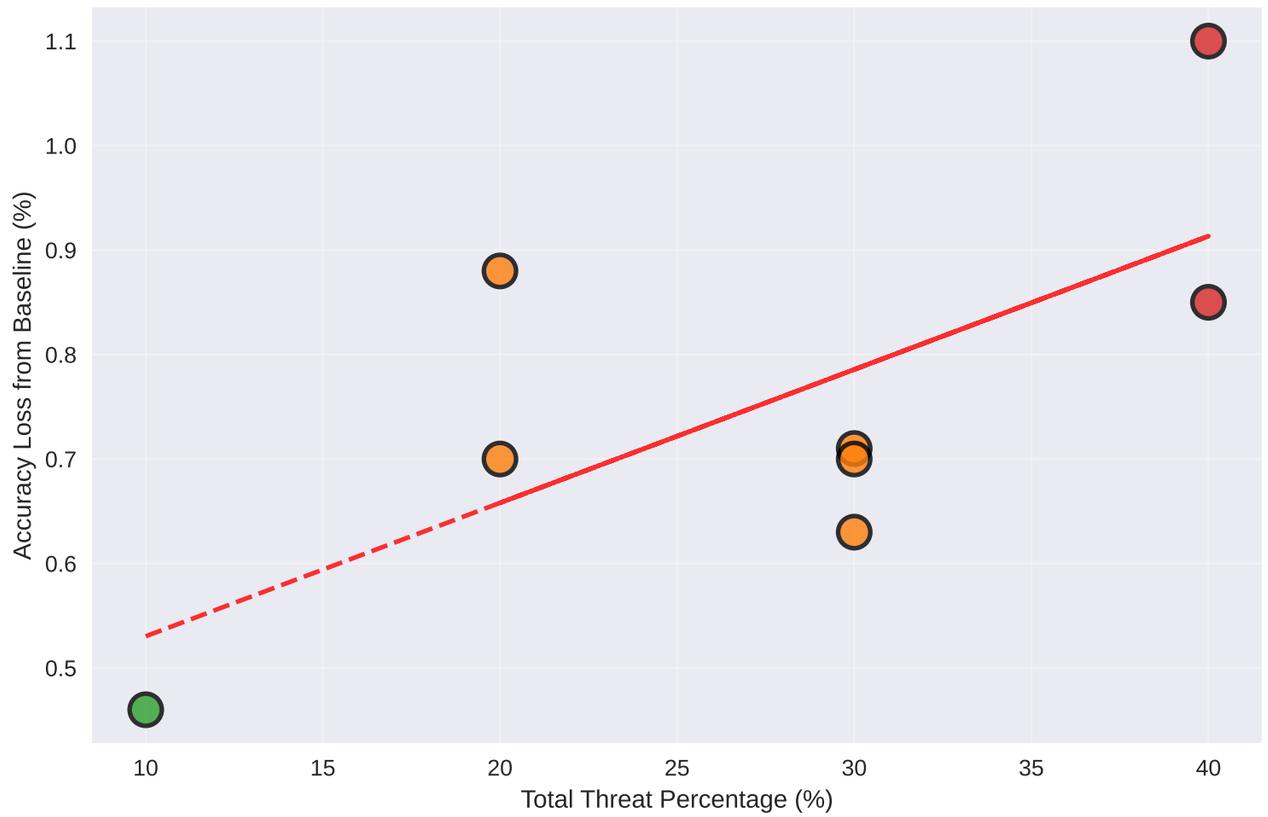


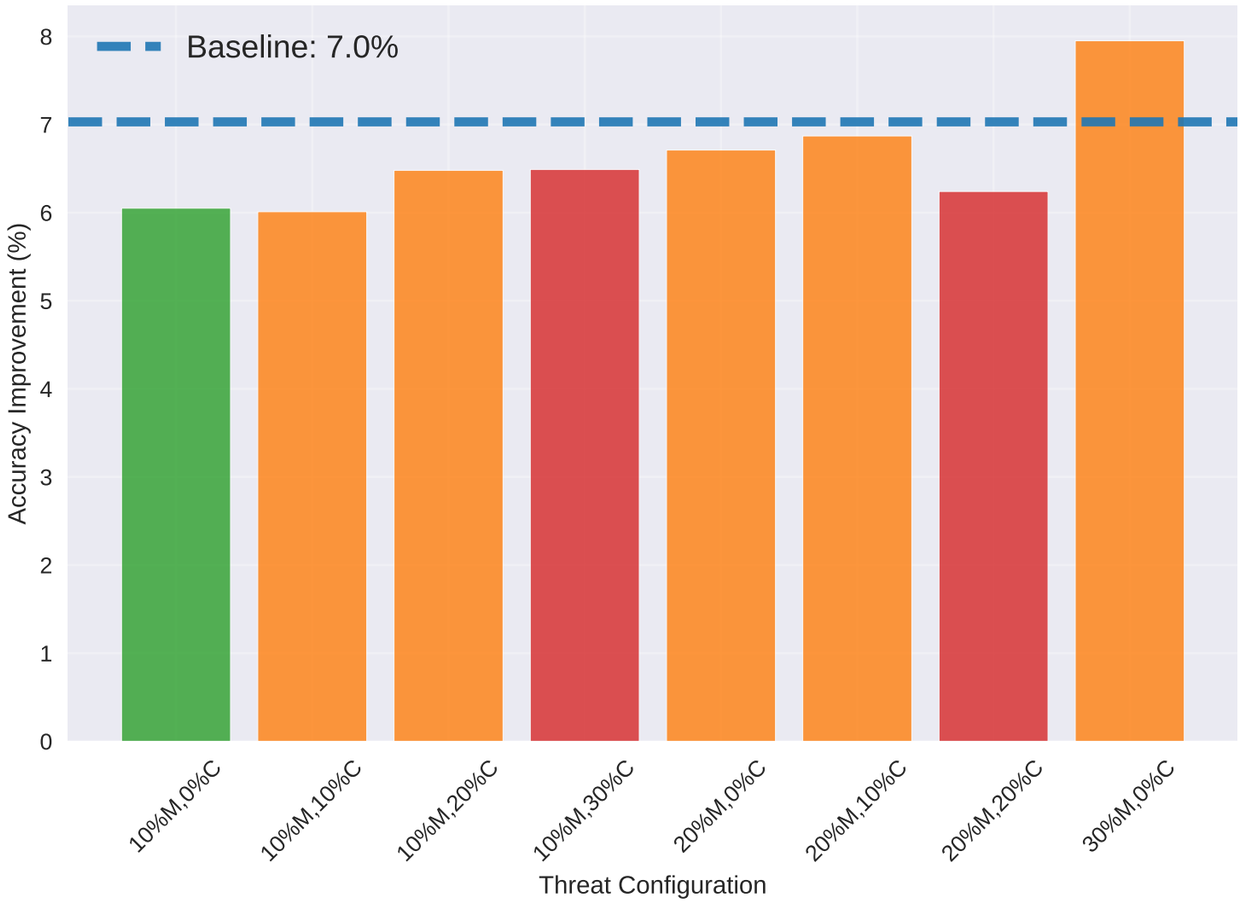


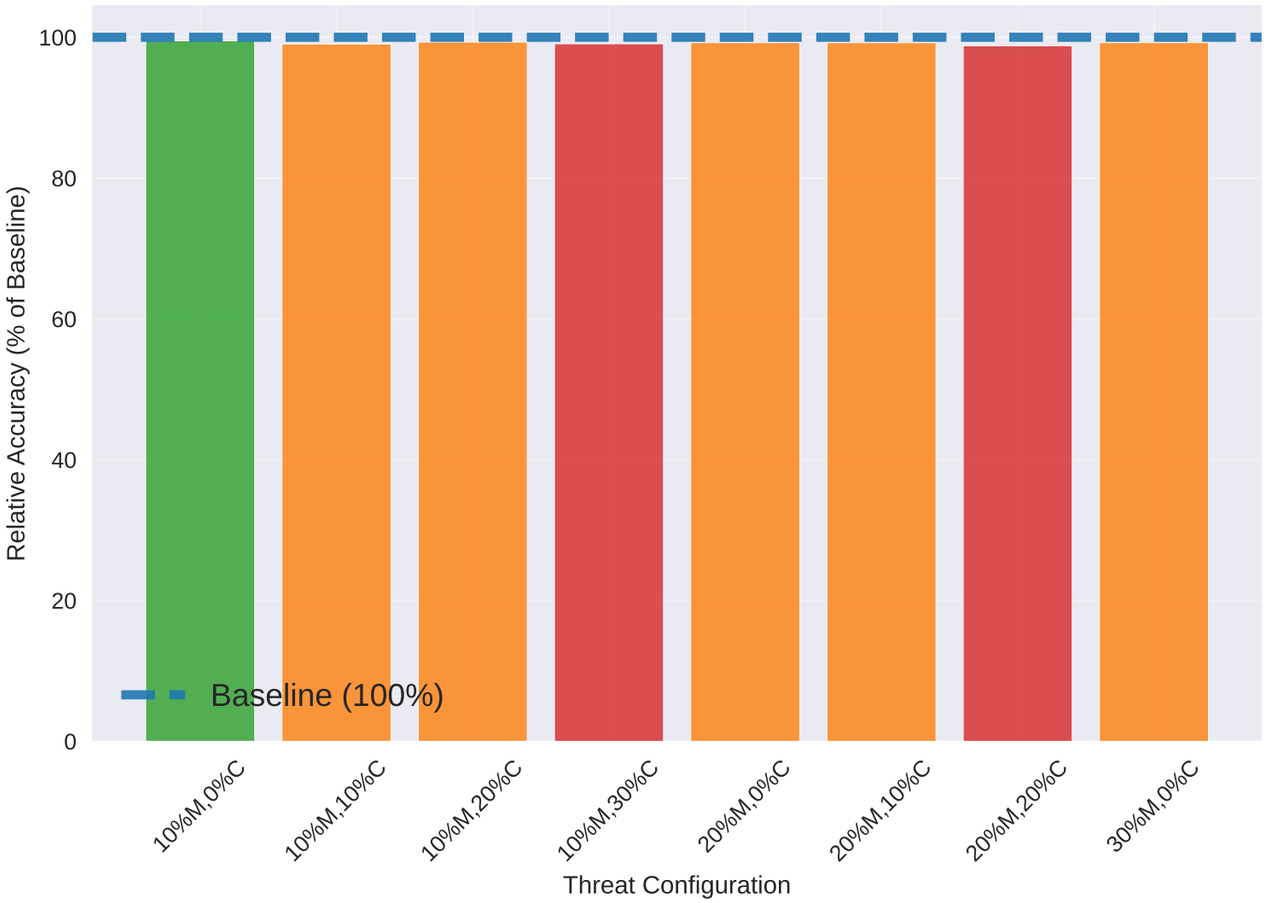


Baseline vs Threats Analysis

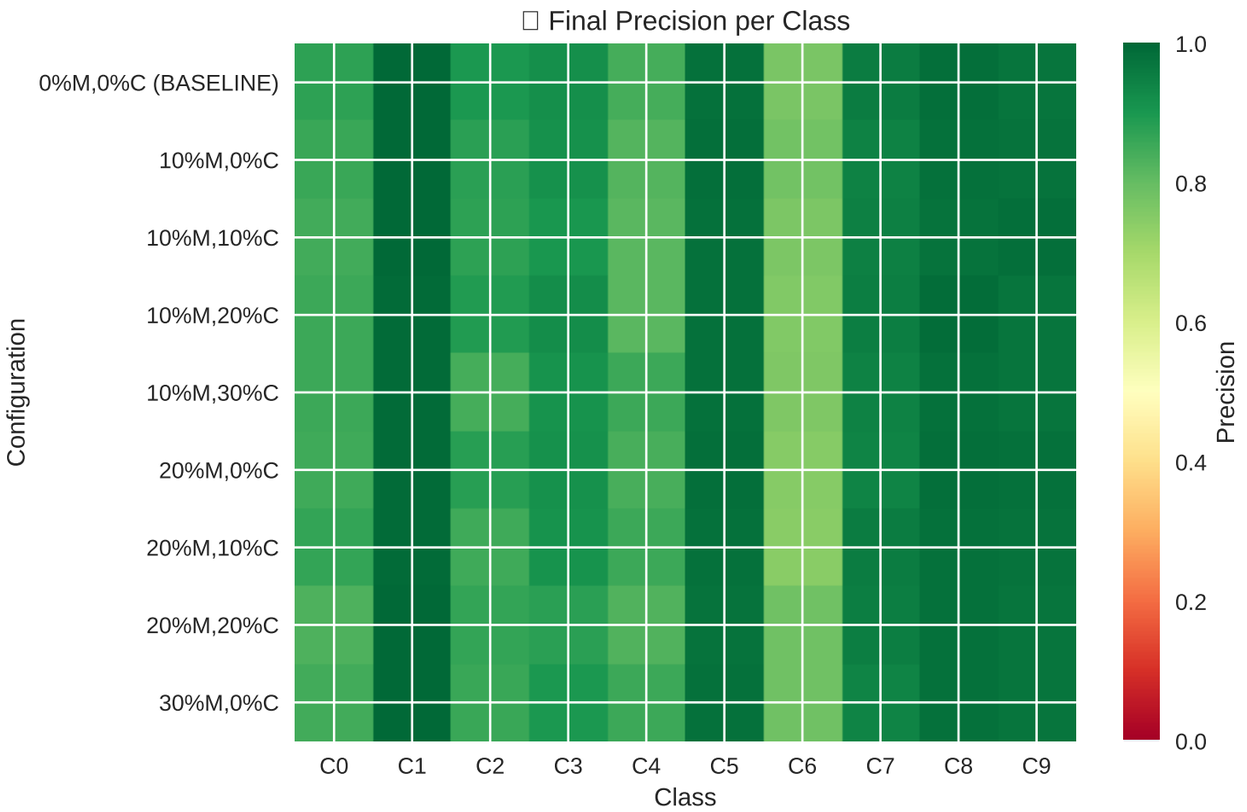


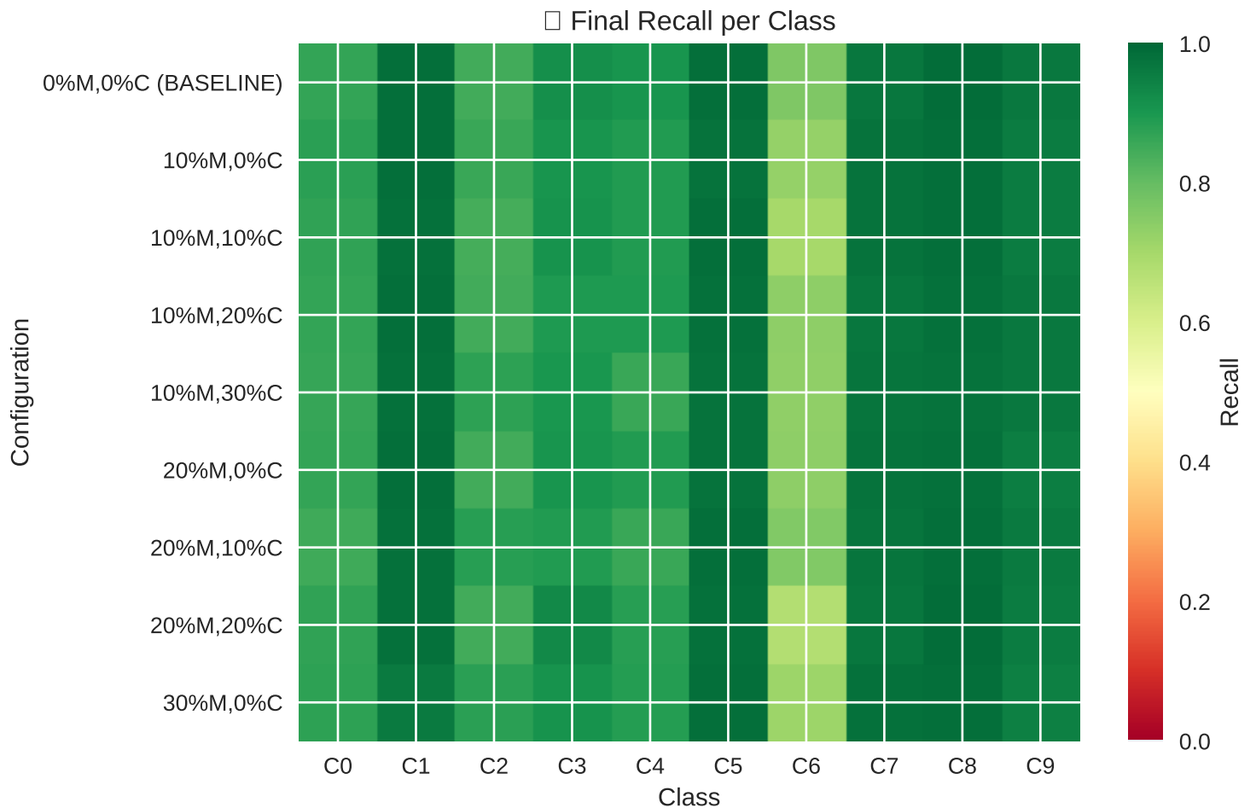


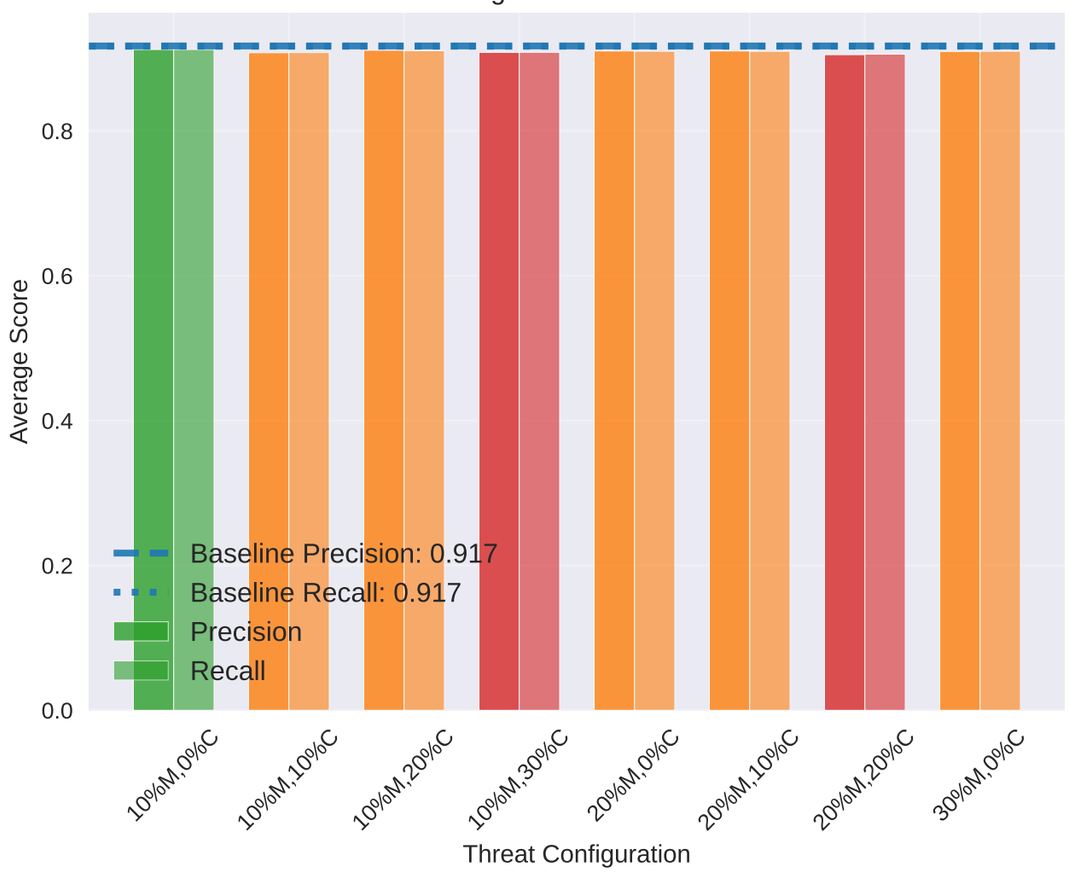


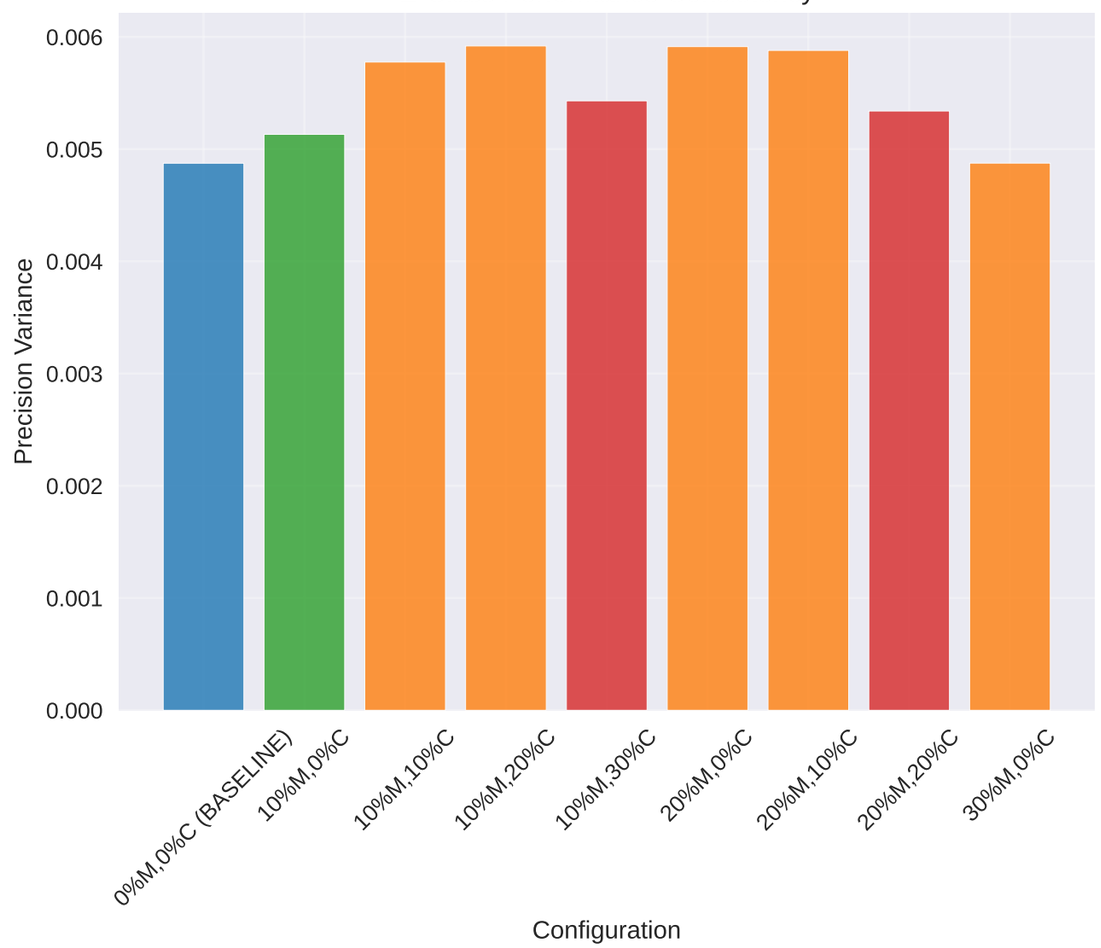


Per-Class Performance Analysis









FL Trust Experiments - Complete Performance Summary

Mal%	Cor%	Tot%	Final Acc	Final Loss	Precision	Recall	F1-Score	Acc Improve	Max Acc	Epochs	Type
0%	0%	0%	91.7%	2.504	0.917	0.917	0.917	+7.0%	91.7%	100	BASELINE
10%	0%	10%	91.2%	2.666	0.912	0.912	0.912	+6.0%	91.4%	100	THREAT
10%	10%	20%	90.8%	2.733	0.908	0.908	0.908	+6.0%	91.3%	100	THREAT
20%	0%	20%	91.0%	2.774	0.910	0.910	0.910	+6.7%	91.4%	100	THREAT
10%	20%	30%	91.1%	2.863	0.911	0.911	0.911	+6.5%	91.5%	100	THREAT
20%	10%	30%	91.0%	2.840	0.910	0.910	0.910	+6.9%	91.1%	100	THREAT
30%	0%	30%	91.0%	2.737	0.910	0.910	0.910	+7.9%	91.2%	100	THREAT
10%	30%	40%	90.8%	3.034	0.908	0.908	0.908	+6.5%	91.0%	100	THREAT
20%	20%	40%	90.6%	3.019	0.905	0.906	0.906	+6.2%	90.9%	100	THREAT

Note: 2 outlier experiment(s) were detected and removed from analysis.
See Outlier Analysis page for details.